ALBAR: ADVERSARIAL LEARNING APPROACH TO MITIGATE BIASES IN ACTION RECOGNITION

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ABSTRACT

Action recognition models frequently exploit spurious static spatial cues — such as background scenes or actor appearance — rather than genuine motion patterns, leading to poor out-of-distribution (OOD) generalization. Prior debiasing approaches have concentrated largely on background bias, often require costly bias-attribute labels or auxiliary critic networks, and rely on benchmarks that suffer from background leakage due to imprecise foreground segmentation. We address both shortcomings simultaneously. First, we introduce **ALBAR** (**A**dversarial **L**earning approach to mitigate **B**iases in **A**ction **R**ecognition), a label-free debiasing framework that pairs each training clip with a static counterpart — constructed by tiling a single middle frame — and penalizes confident predictions on these motion-free inputs through a combination of an adversarial loss, entropy maximization, and a gradient penalty. ALBAR requires no bias annotations, no auxiliary critics, and is compatible with augmentation-based debiasing. Second, we introduce an improved UCF101-SCUBA/SCUFO benchmark that replaces coarse bounding-box masks with tight, per-frame actor segmentations obtained via SAMTrack (Cheng et al., 2023), effectively eliminating background leakage and enabling cleaner evaluation of both background and foreground biases. On HMDB51 (Kuehne et al., 2011), ALBAR achieves a contrasted accuracy of 53.22%, surpassing the prior state of the art by $+12.94$ absolute points. On our improved UCF101 (Soomro et al., 2012) benchmark, ALBAR obtains a contrasted accuracy of 26.23% versus 17.53% for the baseline. Debiasing transfers positively to downstream tasks, improving anomaly detection on UCF-Crime and temporal action localization on THUMOS14.

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1 INTRODUCTION

Deep video models have achieved impressive performance on standard action recognition benchmarks such as UCF101 (Soomro et al., 2012) and HMDB51 (Kuehne et al., 2011). However, these benchmarks are collected from real-world videos that encode strong correlations between action classes and their typical environmental contexts. A model trained on such data can attain high in-distribution accuracy by leveraging *static* spatial cues — the background scene, the actor's clothing, or characteristic object appearances — rather than learning the motion patterns that are truly diagnostic of the action (Geirhos et al., 2020; Weinzaepfel & Rogez, 2021; Li et al., 2018). This shortcut learning strategy fails catastrophically under distribution shift: a model that classifies "skiing" by detecting snow cannot generalize when the background is replaced or removed.

The problem of *background bias* in action recognition has received considerable attention (Choi et al., 2019; Wang et al., 2021; Ding et al., 2022; Duan et al., 2022). Recent work by Li et al. (2023) introduced the SCUBA (Static Cues in the Background) and SCUFO (Static Cues in the Foreground) benchmarks together with the StillMix augmentation, and provided the first comprehensive joint study of background and foreground static bias. Despite this progress, two fundamental limitations persist. First, most debiasing methods either require explicit bias-attribute annotations (Choi et al., 2019) or rely on auxiliary 2D reference networks to identify bias-inducing frames (Li et al., 2023). Second, the UCF101-SCUBA/SCUFO benchmark uses bounding-box-based foreground masks to composite actors onto novel backgrounds, which causes *background leakage*: portions of the original background remain visible around the actor's silhouette, confounding the evaluation.

In this paper we make two complementary contributions to address these limitations.

Improved Benchmark. We reconstruct the UCF101-SCUBA/SCUFO benchmark using precise, per-frame actor segmentation masks obtained by initializing from THUMOS14 bounding boxes and propagating them with SAMTrack (Cheng et al., 2023). Every test video is manually verified to ensure clean foreground separation. We retain three background types from Li et al. (2023) (Places365 (Zhou et al., 2017), VQGAN-CLIP (Crowson et al., 2022), and synthetic sinusoidal textures) and include a conflicting-foreground variant (Confl-FG). The improved segmentation eliminates leakage, producing a more faithful evaluation protocol. Figure 2 illustrates the qualitative difference between the original and improved protocols.

ALBAR Method. We propose ALBAR, an adversarial debiasing framework built on a Swin3D-T encoder (Liu et al., 2022) pretrained on Kinetics-400 (Carreira & Zisserman, 2017). The core idea is simple: for each training video clip, we construct a "static clip" by tiling a single middle frame to match the temporal length of the input. A model relying on static cues will assign confident predictions to these motion-free inputs; a truly motion-grounded model should be maximally uncertain. We therefore optimize three complementary objectives applied to static clips: (i) a *static adversarial loss* $\mathcal{L}_{\text{adv}}$ — the negated cross-entropy — which penalizes confident correct predictions on static inputs; (ii) an *entropy maximization loss* $\mathcal{L}_{\text{ent}}$ that directly encourages uniform output distributions for static clips; and (iii) a *gradient penalty* $\mathcal{L}_{\text{gp}}$ on the model's output with respect to the static input for training stability. The full objective requires no bias-attribute labels, no auxiliary critics, and is compatible with existing augmentation-based debiasing such as StillMix (Li et al., 2023).

We evaluate ALBAR on HMDB51 and our improved UCF101 benchmark, and find large improvements over the prior state of the art in contrasted accuracy with minimal IID accuracy drop. We further demonstrate that the debiased representations transfer to downstream tasks, improving anomaly detection AUC on UCF-Crime (Sultani et al., 2018) and mean AP on THUMOS14 (Jiang et al., 2014) temporal action localization.

2 Related Work

Static Bias in Action Recognition. Early work demonstrated that action classifiers exploit dataset-level correlations between scenes and action classes (Li et al., 2018; Li & Vasconcelos, 2019). Huang et al. (2018) showed that many actions can be recognized from single frames, suggesting that temporal reasoning is underutilized. Weinzaepfel & Rogez (2021) introduced the Mimetics dataset of out-of-context mimed actions to probe this limitation. Byvshev et al. (2022) found that 3D convolutional networks are inherently biased towards appearance, not motion.

Debiasing Methods. Choi et al. (2019) (SDN) used an adversarial scene-type loss and a human-mask confusion loss to reduce background dependence; however, their approach requires scene labels. Wang et al. (2021) (Background Erasing, BE) proposed adding a static frame to all positions of a video during self-supervised pretraining. Ding et al. (2022) (FAME) merges foreground regions across videos during contrastive learning. Duan et al. (2022) (SMAD) combines multi-aspect adversarial training with a spatial actionness reweighting module for open-set settings. Li et al. (2023) (StillMix) identifies bias-inducing frames using a 2D reference network and mixes them with training videos; they also introduced SCUBA/SCUFO as evaluation benchmarks. Our ALBAR differs from all of these in that it requires no bias-attribute annotations, no auxiliary critics, and no 2D reference networks — making it simpler to deploy in practice.

Adversarial Debiasing. Adversarial learning has been widely used for debiasing in the fairness literature (Zhang et al., 2018; Goodfellow et al., 2014). Approaches such as Bahng et al. (2020) train a debiased representation by encouraging it to differ from biased representations. In the video domain, Choi et al. (2019) applies an adversarial scene-type loss, and Zhai et al. (2023) proposes adversarial scene reconstruction for open-set recognition. Our approach uses adversarial loss directly on the classification head rather than on an auxiliary representation, which simplifies training and avoids the instability of critic-based methods. The gradient penalty we adopt is inspired by Wasserstein GAN training (Gulrajani et al., 2017).

Video Transformers. The Video Swin Transformer (Liu et al., 2022) introduced locality-biased spatio-temporal attention by adapting the Swin architecture to video, achieving strong performance

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on Kinetics-400/600 with efficient computation. We use the Swin3D-Tiny variant as our backbone due to its favorable accuracy-efficiency trade-off.

3 IMPROVED UCF101-SCUBA/SCUFO BENCHMARK

3.1 BACKGROUND AND MOTIVATION

The SCUBA and SCUFO benchmarks introduced by Li et al. (2023) provide a principled protocol for evaluating static bias in action recognition:

- **SCUBA** (Static Cues in the Background): The original actor is composited onto novel backgrounds. A model relying on background cues should fail; a model using motion should succeed.
- **SCUFO** (Static Cues in the Foreground): A single frame is tiled to form a static clip that preserves appearance without motion. A model relying on foreground appearance (clothing, body shape) will succeed; a true motion model should fail.
- **Confl-FG**: A static foreground frame from one action class is composited onto backgrounds from another class, creating conflicting signals.

The original UCF101-SCUBA/SCUFO benchmark uses bounding-box masks to isolate actors, resulting in background *leakage*: snow, grass, or other scene-specific pixels remain visible around the actor's silhouette. As shown in Figure 2(a), this leakage allows a biased model to exploit residual background information even in the SCUBA protocol, confounding the evaluation.

3.2 SEGMENTATION-BASED RECONSTRUCTION

We reconstruct the benchmark using tight, per-frame actor segmentation. Our pipeline is as follows:

Initialization. For each UCF101 test video, we initialize actor masks using the corresponding THUMOS14 bounding box annotations.

Mask propagation. We apply SAMTrack (Cheng et al., 2023), which combines the Segment Anything Model (Kirillov et al., 2023) with DeAOT (Yang & Yang, 2022), to propagate fine-grained instance masks across all frames of each video.

Manual verification. Every test video is manually reviewed to confirm accurate foreground separation and to remove any remaining leakage cases.

Protocol synthesis. SCUBA clips are created by compositing the original actor mask onto novel backgrounds; SCUFO clips are created by selecting a SCUBA frame and tiling it to match the clip length, eliminating all temporal motion. We use three background types: Places365 test set images (Zhou et al., 2017), VQGAN-CLIP generated textures (Crowson et al., 2022), and synthetic sinusoidal stripe patterns.

As shown in Figure 2(b), the improved segmentation removes the background pixels that leak through the bounding-box mask, yielding a cleaner evaluation.

3.3 EVALUATION METRICS

We report Top-1 accuracy on: (i) the IID test set; (ii) average SCUBA accuracy across background types (↑); (iii) average SCUFO accuracy (↓, lower is better as it indicates less reliance on static foreground cues); (iv) Confl-FG accuracy (↑); and (v) **contrasted accuracy** (↑): the fraction of video pairs where the model correctly classifies the SCUBA clip while simultaneously misclassifying the paired SCUFO clip. This metric jointly rewards robustness to background changes and insensitivity to static foreground cues, making it the primary metric of interest.

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4 METHOD: ALBAR

4.1 ARCHITECTURE AND STATIC CLIP CONSTRUCTION

ALBAR uses a Video Swin Transformer (Swin3D-T) (Liu et al., 2022) pretrained on Kinetics-400 (Carreira & Zisserman, 2017) as its feature encoder, followed by a linear classification head. Given a batch of motion clips $\{\mathbf{x}_t^{(i)}\}_{i=1}^B$, where each clip consists of T frames sampled from a training video, we construct a corresponding batch of *static clips* $\{\bar{\mathbf{x}}_{\bar{t}}^{(i)}\}_{i=1}^B$ by selecting the middle frame of $\mathbf{x}_t^{(i)}$ and tiling it T times. This construction ensures that the static clip shares the same foreground appearance as the motion clip while containing zero temporal variation. The architecture is illustrated in Figure 1.

4.2 TRAINING OBJECTIVES

The total training loss is:

$$\mathcal{L} = \lambda_t \, \mathcal{L}_{\text{CE}}(\mathbf{x}_t, y) \; - \; \lambda_s \, \mathcal{L}_{\text{adv}}(\bar{\mathbf{x}}_{\bar{t}}, y) \; + \; \lambda_e \, \mathcal{L}_{\text{ent}}(\bar{\mathbf{x}}_{\bar{t}}) \; + \; \lambda_{gp} \, \mathcal{L}_{\text{gp}}(\bar{\mathbf{x}}_{\bar{t}}), \tag{1}$$

where y is the ground-truth label, λ_t=1.0, λ_s=1.0, λ_e=4.25, λ_{gp}=10.0 are scalar weights.

Cross-Entropy Loss on Motion Clips ($\mathcal{L}_{\text{CE}}$). The standard supervised objective ensures the model learns to classify the actual motion clips correctly:

$$\mathcal{L}_{\text{CE}}(\mathbf{x}_t, y) = -\log p_y(f(\mathbf{x}_t)), \tag{2}$$

where f denotes the model and p_y the probability assigned to class y.

Static Adversarial Loss ($\mathcal{L}_{\text{adv}}$). The adversarial term penalizes high-confidence predictions on static clips by *negating* the cross-entropy:

$$\mathcal{L}_{\text{adv}}(\bar{\mathbf{x}}_{\bar{t}}, y) = \mathcal{L}_{\text{CE}}(\bar{\mathbf{x}}_{\bar{t}}, y) = -\log p_y(f(\bar{\mathbf{x}}_{\bar{t}})). \tag{3}$$

Since $\mathcal{L}_{\text{adv}}$ enters the total loss with a negative sign (Eq. 1), minimizing $\mathcal{L}$ maximizes the cross-entropy on static clips — directly penalizing correct predictions made without motion. Crucially, this formulation does not use gradient reversal layers; it directly maximizes the cross-entropy at the output of the classification head.

Entropy Maximization Loss ($\mathcal{L}_{\text{ent}}$). To further enforce that static clips produce uninformative predictions, we add an entropy maximization term. We define:

$$\mathcal{L}_{\text{ent}}(\bar{\mathbf{x}}_{\bar{t}}) = \mathbb{E}\left[\sum_c p_c(f(\bar{\mathbf{x}}_{\bar{t}})) \log p_c(f(\bar{\mathbf{x}}_{\bar{t}}))\right], \tag{4}$$

which is the *negative* entropy of the output distribution. Adding $+\lambda_e \, \mathcal{L}_{\text{ent}}$ to the total loss (which is minimized) is equivalent to maximizing the entropy, driving the model toward uniform predictions on static clips.

Gradient Penalty ($\mathcal{L}_{\text{gp}}$). Inspired by Wasserstein GAN training (Gulrajani et al., 2017), we include a gradient penalty on the model's output logits with respect to the static input:

$$\mathcal{L}_{\text{gp}}(\bar{\mathbf{x}}_{\bar{t}}) = \left\| \nabla_{\bar{\mathbf{x}}_{\bar{t}}} f(\bar{\mathbf{x}}_{\bar{t}}) \right\|_2, \tag{5}$$

computed as the L2 norm of the gradient of the output with respect to the static clip. This regularizer smooths the loss landscape and improves training stability.

4.3 COMPATIBILITY WITH DATA AUGMENTATION

ALBAR can be combined with existing augmentation-based debiasing. In our experiments we pair it with StillMix (Li et al., 2023), which mixes still frames from other videos into training clips. Since StillMix acts on the training data distribution while ALBAR acts on the model's response to static inputs, the two approaches are complementary and their combination yields further improvements over either alone.

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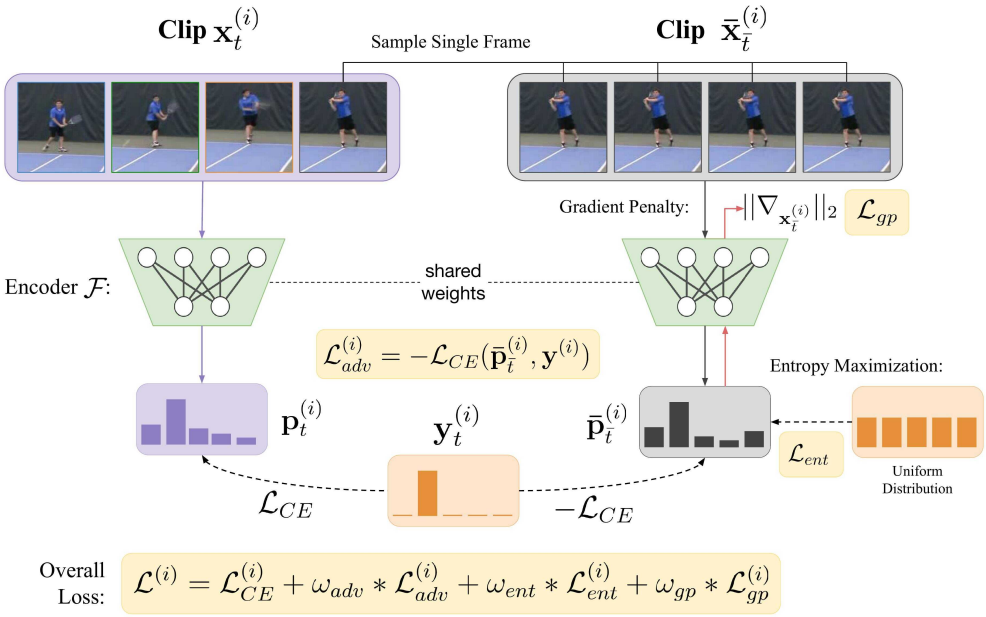

Figure 1: Overview of ALBAR. Given a motion clip $\mathbf{x}_t^{(i)}$, we sample the middle frame and tile it to create a static clip $\bar{\mathbf{x}}_{\bar{t}}^{(i)}$. The same Swin3D-T encoder processes both clips. The motion clip contributes a standard cross-entropy loss; the static clip receives an adversarial (negated cross-entropy) loss, an entropy maximization loss, and a gradient penalty to discourage reliance on static spatial cues.

(a) Original protocol: background leakage (snow visible around actor).

(b) Improved protocol: tight segmentation eliminates leakage.

Figure 2: Example frames from UCF101-SCUBA-Sinusoid for the class "Skiing". (a) The original protocol uses bounding-box masks, leaving snow visible in the background. (b) Our improved protocol uses SAMTrack segmentation to produce tight actor masks, cleanly separating foreground from the novel background.

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5 Experiments

5.1 Experimental Setup

Backbone. All experiments use a Video Swin Transformer (Swin3D-Tiny) (Liu et al., 2022) pre-trained on Kinetics-400 (Carreira & Zisserman, 2017). Input clips consist of 32 frames at 224×224 resolution with temporal stride 2.

Training. We fine-tune with AdamW (Loshchilov & Hutter, 2017) (lr $= 10^{-5}$, weight decay 5×10^{-2}) for 50 epochs with cosine learning rate scheduling. Loss weights are λ_t=1.0, λ_s=1.0, λ_e=4.25, λ_{gp}=10.0.

Datasets. We evaluate on HMDB51 (Kuehne et al., 2011) (51 classes) and UCF101 (Soomro et al., 2012) (101 classes) using both IID test splits and the OOD SCUBA/SCUFO protocols described in Section 3.

Baselines. We compare against: (i) training with no debiasing; (ii) Mixup (Zhang et al., 2017); (iii) VideoMix (Yun et al., 2020); (iv) SDN (Choi et al., 2019); (v) Background Erasing (BE) (Wang et al.,

Table 1: Comparison results on IID and OOD test sets of various debiasing methods on HMDB51. All experiments use Swin-T pretrained using Kinetics-400. The light blue column highlights the contrasted accuracy.

Augmentation or Debiasing	IID	OOD			
		Avg SCUBA ↑	Avg SCUFO ↓	Confl-FG ↑	Contra. Acc. ↑
None	73.92	43.93	20.46	36.58	27.84
$\text{Mixup}_{\text{ICLR'18}}$	74.58	43.10	21.17	36.62	26.09
$\text{VideoMix}_{\text{arXiv'20}}$	73.31	39.39	20.44	32.68	23.13
$\text{SDN}_{\text{NeurIPS'19}}$	74.66	40.02	20.22	34.87	22.88
$\text{BE}_{\text{CVPR'21}}$	74.31	43.56	19.96	35.99	27.84
$\text{ActorCutMix}_{\text{CVIU'23}}$	74.05	46.79	22.07	36.97	28.12
$\text{FAME}_{\text{CVPR'22}}$	73.79	51.40	26.92	39.61	29.66
$\text{StillMix}_{\text{ICCV'23}}$	**74.82**	51.81	13.39	47.38	40.28
Ours	72.81	**53.53** ↑21.9%	**1.50** ↓92.7%	**48.13** ↑31.4%	**53.22** ↑91.2%
Ours w/ StillMix aug.	74.31	54.24	1.35	47.07	53.68

2021); (vi) ActorCutMix (Zou et al., 2023); (vii) FAME (Ding et al., 2022); and (viii) StillMix (Li et al., 2023).

5.2 Main Results on HMDB51

Table 1 reports results on HMDB51. ALBAR achieves:

- **Contrasted Accuracy: 53.22%** vs. 40.28% for StillMix (**+12.94** absolute, **+32.1%** relative).
- **Avg SCUBA: 53.53%** (**+21.9%** relative improvement over StillMix).
- **Avg SCUFO: 1.50%** (**−92.7%** reduction), indicating very low reliance on foreground static appearance.
- **Confl-FG: 48.13%** (**+31.4%** relative improvement).

The IID accuracy is 72.81%, a modest drop of $\approx 1.0\%$ from the 73.92% baseline — a favorable trade-off given the large OOD gains. Combining ALBAR with StillMix augmentation further improves contrasted accuracy to 53.68% while recovering the IID accuracy to 74.31%.

5.3 Main Results on UCF101 (Improved Protocol)

Table 2 shows results on our improved UCF101 benchmark. The improved segmentation significantly increases the difficulty: the baseline contrasted accuracy drops from the HMDB51 counterpart, and even strong baselines struggle. ALBAR achieves:

- **Contrasted Accuracy: 26.23%** vs. 17.53% baseline (**+49.6%** relative).
- **Avg SCUBA: 26.25%** (**+39.8%** relative).
- **Avg SCUFO: 0.14%** (**−90.7%**), the lowest among all methods, demonstrating that ALBAR almost entirely suppresses foreground static bias on this benchmark.
- **Confl-FG: 54.02%** (**+9.4%** relative).

Combining with StillMix further boosts contrasted accuracy to 31.14% and Avg SCUBA to 31.17%.

5.4 Ablation Study: Loss Components

Table 3 presents an ablation on HMDB51 where we systematically add loss components. Row (a) is the standard cross-entropy baseline. Adding $\mathcal{L}_{\text{adv}}$ alone (b) effectively eliminates SCUFO accuracy to 0.00% and raises contrasted accuracy to 42.29%, confirming that negating the cross-entropy on static

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Table 2: Results on IID and OOD test sets of various debiasing methods on UCF101 using our proposed updated protocol. The light blue column highlights the contrasted accuracy. All experiments use Swin-T pretrained using Kinetics-400.

Augmentation or Debiasing	IID	OOD			
		Avg SCUBA ↑	Avg SCUFO ↓	Confl-FG ↑	Contra. Acc. ↑
None	95.51	18.78	1.50	49.36	17.53
StillMix ICCV'23	**96.14**	24.68	0.36	**55.03**	24.40
Ours	94.98	**26.25** ↑39.8%	**0.14** ↓90.7%	54.02 ↑9.4%	**26.23** ↑49.6%
Ours w/ StillMix aug.	94.71	31.17	0.16	58.84	31.14

clips is a powerful debiasing signal. Adding $\mathcal{L}_{\text{ent}}$ alone (c) achieves 44.30% contrasted accuracy. The gradient penalty alone (d) provides modest improvement (29.24%). Combining $\mathcal{L}_{\text{adv}}$ + $\mathcal{L}_{\text{ent}}$ (e) reaches 46.72% contrasted accuracy. The full model with all four components (h) achieves the best contrasted accuracy (53.22%) and Avg SCUBA (53.53%), confirming that each component contributes complementary information.

Table 3: Ablation to determine the effectiveness of each component loss objective.

	$\mathcal{L}_{CE}$	$\mathcal{L}_{adv}$	$\mathcal{L}_{ent}$	$\mathcal{L}_{gp}$	IID	OOD			
						Avg SCUBA ↑	Avg SCUFO ↓	Confl-FG ↑	Contra. Acc. ↑
(a)	✓	✗	✗	✗	73.92	43.93	20.46	36.58	27.84
(b)	✓	✓	✗	✗	72.61	42.29	**0.00**	32.77	42.29
(c)	✓	✗	✓	✗	71.57	46.02	2.97	38.40	44.30
(d)	✓	✗	✗	✓	72.09	41.13	14.69	35.16	29.24
(e)	✓	✓	✓	✗	72.22	46.81	0.49	41.13	46.72
(f)	✓	✓	✗	✓	71.70	45.92	**0.00**	38.52	45.92
(g)	✓	✗	✓	✓	**73.86**	49.14	9.74	42.89	41.11
(h)	✓	✓	✓	✓	72.81	**53.53**	1.50	**48.13**	**53.22**

5.5 Ablation Study: Static Frame Sampling Strategy

Table 4 evaluates different strategies for choosing which frame to tile into the static clip. We compare sampling the first, middle, last, or a random frame. The middle-frame strategy achieves the best contrasted accuracy (53.22%) tied with random sampling (53.02%). The middle frame is the most representative of the clip's content (the actor is typically well-centered), whereas first/last frames may capture transitional poses. We adopt the middle frame as our default for its stable, deterministic nature.

Table 4: Ablation to evaluate static frame sampling strategy.

Frame Sampling Strategy	IID	OOD			
		Avg SCUBA ↑	Avg SCUFO ↓	Confl-FG ↑	Contra. Acc. ↑
Random	73.20	53.22	0.42	49.84	53.02
First	72.75	50.92	0.18	45.59	50.91
Middle	72.81	53.53	1.50	48.13	**53.22**
Last	72.68	49.49	0.40	42.42	49.40

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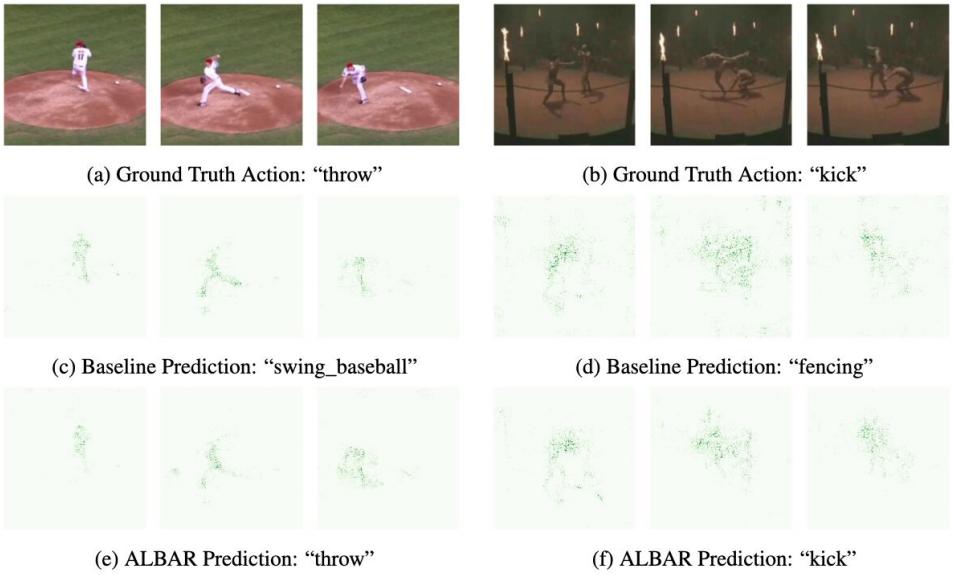

(a) Ground Truth Action: “throw” (b) Ground Truth Action: “kick”

(c) Baseline Prediction: “swing_baseball” (d) Baseline Prediction: “fencing”

(e) ALBAR Prediction: “throw” (f) ALBAR Prediction: “kick”

Figure 3: Qualitative examples from the HMDB51 test set. The baseline model (top) assigns incorrect labels by relying on static background or context cues. ALBAR (bottom) produces correct predictions by focusing on motion features.

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5.6 Downstream Task Transfer

To assess whether debiasing improves generalization beyond the benchmark test sets, we evaluate the ALBAR-trained Swin3D-T encoder on two downstream tasks (Table 5):

Anomaly Detection (UCF-Crime (Sultani et al., 2018)). Weakly supervised anomaly detection benefits from motion-grounded representations that do not conflate unusual static scenes with anomalous events. ALBAR achieves an AUC of $\mathbf{84.91}$ vs. 82.39 for the baseline (**+2.52** points), evaluated following Chen et al. (2023).

Temporal Action Localization (THUMOS14 (Jiang et al., 2014)). Temporal localization requires identifying the precise start and end of an action in an untrimmed video — a task that benefits from avoiding background shortcuts. ALBAR achieves a mean AP of $\mathbf{55.20}$ vs. 54.89 for the baseline (**+0.31** mAP), evaluated following Shi et al. (2023).

Table 5: Evaluation of our debiased model across downstream anomaly detection (UCF_Crime) and temporal action localization (THUMOS14) tasks.

Method	Action Recognition HMDB51 (OOD)	Anomaly Detection UCF_Crime	Temporal Action Localization THUMOS14
Baseline	27.84	82.39	54.89
Ours	**53.22**	**84.91**	**55.20**

5.7 Qualitative Analysis

Figure 3 shows examples from the HMDB51 test set where the baseline model fails by exploiting static context. For instance, the baseline predicts “golf” for a “baseball” clip because the grass background is associated with golf-like settings; ALBAR correctly predicts “baseball” by focusing on the actor’s motion. These examples illustrate that ALBAR successfully shifts the model’s attention toward motion features.

5.8 Comparison with Prior Adversarial Debiasing

Unlike Choi et al. (2019), which requires scene-type labels and a separate adversarial critic network, ALBAR is self-supervised in the sense that the static clips are automatically derived from the training data without any additional annotations. The adversarial signal operates at the level of the classification head itself, avoiding the instability common in GAN-style minimax training. Unlike Wang et al. (2021), which modifies the contrastive loss for self-supervised pretraining, ALBAR is a fine-tuning-stage method applicable on top of any pretrained model.

6 Discussion

Why does entropy maximization complement adversarial loss? The adversarial loss $-\mathcal{L}_{\text{CE}}$ penalizes correct predictions on static clips but does not explicitly promote uniform distributions — the model might simply assign high probability to an incorrect class, which would also reduce $\mathcal{L}_{\text{CE}}$. The entropy maximization term $\mathcal{L}_{\text{ent}}$ fills this gap by explicitly pushing the output distribution toward uniform, regardless of which class receives the highest probability. The two losses thus target different failure modes and are complementary.

Role of the gradient penalty. The gradient penalty penalizes the sensitivity of the output to small changes in the static input, discouraging the model from relying on fine-grained spatial details of the static frame. This acts as a regularizer that smooths the optimization landscape and prevents the model from developing unstable oscillatory behaviors when the adversarial and entropy losses are simultaneously active.

Benchmark improvements and impact. Our segmentation-based benchmark reveals that the original UCF101-SCUBA/SCUFO results were inflated by leakage. The improved benchmark is substantially harder (baseline contrasted accuracy drops to 17.53% on UCF101), providing a more faithful test of static-bias elimination. We expect this benchmark to be a more reliable tool for future debiasing research.

Limitations. ALBAR incurs a modest IID accuracy drop (≈ 0.5–1.2%) compared to unregularized fine-tuning, which may be unacceptable in highly accuracy-sensitive applications. The static clip construction via middle-frame tiling is a coarse approximation; future work could explore more sophisticated spatial-only clips (e.g., optical-flow-nullified videos). The contrasted accuracy metric, while informative, does not capture all aspects of bias: a model could achieve high contrasted accuracy while still using appearance cues that happen to agree with the correct label.

7 Conclusion

We presented ALBAR, an adversarial learning framework that mitigates static spatial bias in video action recognition without requiring bias-attribute labels or auxiliary critic networks. By penalizing confident predictions on automatically constructed static (motion-free) clips through a combination of adversarial loss, entropy maximization, and gradient penalty, ALBAR forces the model to rely on genuine motion features. We also introduced an improved UCF101-SCUBA/SCUFO benchmark with tight, segmentation-based actor masks that eliminate background leakage and enable cleaner evaluation. ALBAR sets a new state of the art in contrasted accuracy on both HMDB51 (53.22%, $+12.94$ vs. prior best) and the improved UCF101 benchmark (26.23%, $+8.70$ vs. baseline), and the debiased representations transfer positively to downstream tasks. We hope that ALBAR and the improved benchmark will serve as useful tools for the community to develop and evaluate action recognition models that reason about motion rather than exploiting static shortcuts.

Acknowledgments

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